

Le (Leonard) Zhang

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Education

Ph.D. in Computer Science and Engineering , University of Michigan, Ann Arbor	Expected May 2029
Advisor: Professor Ke Sun	
M.S. in Computer Science , University of California, San Diego	June 2025
Advisor: Professor Tajana Šimunić Rosing	
B.A. in Computer Science , University of North Carolina at Chapel Hill	December 2022
Major in Computer Science and Minor in Mathematics	
Advisor: Professor Shahriar Nirjon	

Research Interests

- Resource-efficient machine learning and embedded intelligences.
- Energy-harvesting, batteryless systems and intermittent computing.
- Sustainable and wearable human-centered designs and emerging technologies.
- Intelligent acoustic and speech applications.

Awards & Grants

Dean's List: University of North Carolina at Chapel Hill	2019—2022
Honorable Mentioned Award: ICPC Mid-Atlantic Regional	2019

Conference Publications

1. **Le Zhang***, Quanling Zhao*, Run Wang, Shirley Bian, Onat Gungor, Flavio Ponzina, and Tajana Rosing. "Offload Rethinking by Cloud Assistance for Efficient Environmental Sound Recognition on LPWANs." *The ACM Conference on Embedded Networked Sensor Systems (SenSys)*, 2025. (Core Rank A*, Full Paper, Acceptance Rate: 19.2%)
2. **Le Zhang**, Onat Gungor, Flavio Ponzina, and Tajana Rosing. "E-QUARTIC: Energy Efficient Edge Ensemble of Convolutional Neural Networks for Resource-Optimized Learning." *The Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025. (Core Rank C, Acceptance Rate: 31%)
3. Yubo Luo, **Le Zhang**, Zhenyu Wang, and Shahriar Nirjon. "Antler: Exploiting Task Affinity for Efficient Multitask Learning on Low-Resource Systems." *The International Conference on Embedded Wireless Systems and Networks (EWSN)*, 2024. (Core Rank B, Acceptance Rate: 22.8%)

Short Paper & Abstracts

1. Run Wang*, Shirley Bian*, Xiaofan Yu, Quanling Zhao, **Le Zhang**, and Tajana Rosing. "Poster: Resource-Efficient Environmental Sound Classification Using Hyperdimensional Computing." *The ACM Conference on Embedded Networked Sensor Systems (SenSys)*, 2024. (Core Rank A*)
2. **Le Zhang**, Yubo Luo, and Shahriar Nirjon. "Demo Abstract: Capuchin: A Neural Network Model Generator for 16-bit Microcontrollers." *The ACM/IEEE Conference on Information Processing in Sensor Networks (IPSN)*, 2022. (Core Rank A*)

Research Experience

Student Researcher , System Energy Efficiency Lab (SeeLab), UC San Diego	2024
Resource-Constrained Collaborative Environmental Sound Recognition over LPWANs.	

Advisor: Prof. Tajana Rosing

- Proposed a novel cloud-assisted offloading for resource-efficient learning on low-power wide-area networks (LPWANs) like LoRa. Designed for adaptive collaboration for resource-constrained energy-harvesting systems and low-bit rate, unreliable wireless channels.
- Achieved 2.5-12.5% improvement in accuracy, 80x in energy savings, and 220x in latency reduction compared to the state-of-the-art methods.
- 12-page full paper accepted by *SenSys 2025*.

Student Researcher, the Systems Energy Efficiency Lab (SeeLab), UC San Diego

2023—2024

Energy Efficient Ensemble Learning on Energy-Harvesting Systems.

Advisor: Prof. Tajana Rosing

- Developed an energy-adaptive ensemble learning framework for efficient inference and training on STM32 microcontroller. Improved the reliability of energy-harvesting machine learning system in low-energy conditions up to 40%.
- Paper published in *ASP-DAC 2025*.

Undergraduate Research Assistant, Embedded Intelligence Lab, UNC-Chapel Hill

2021—2022

Exploiting Task Affinity for Efficient Multitask Learning on Low-Resource Systems.

Advisor: Prof. Shahriar Nirjon

- Developed and evaluated an efficient multitask learning framework and memory management algorithm to optimize the orders of multitask executions on low-resource embedded systems.
- Paper accepted by *EWSN 2024*.

Undergraduate Research Assistant, Embedded Intelligence Lab, UNC-Chapel Hill

2021—2022

Efficient Neural Network Implementations on 16-bit Microcontrollers.

Advisor: Prof. Shahriar Nirjon

- Developed a neural network model generator for 16-bit TI MSP430 series microcontrollers. Improved workflow from minutes to several seconds.
- Reduced more than 50% inference time and energy consumption using hardware accelerator compared to the state-of-the-arts.
- Demo published in *IPSN 2022*.

Mentored Research, UNC-Chapel Hill

2021

Remote Collaborative Physics Simulation for High School Physics Education.

Advisor: Prof. Prasun Dewan

- Developed a remote user interface coupling platform for physics simulations in high school physics education using RPCs and IPCs to synchronize physical animations remotely.

Teaching & Mentoring

Research Mentoring, Energy-harvesting Machine Learning Testbed

Fall 2024

Mentee: Run Wang, undergraduate student in ECE at UCSD. Shirley Bian, undergraduate student in CS at UCSD.

Research Mentoring, Hyperdimensional Computing on Embedded Systems

Spring 2024

Mentee: Run Wang, undergraduate student in ECE at UCSD. Publication in *SenSys '24* poster.

Teaching Assistant, COMP 301: Foundations of Programming, UNC-Chapel Hill

Summer 2021

Assisted in course instruction under Prof. Prasun Dewan, helping students in introductory-level Java programming.